

1-20 A constant current of 2 A charges a battery for 4 hours. During the charging period, the battery voltage is $v(t) = 12 - 2e^{-t}$, where t is in hours. Determine the energy stored in the battery.

1-22 Figure P1-22 shows an electric circuit with a voltage and a current variable assigned to each of the six devices. The device voltages and currents are observed to be

	$v(\text{V})$	$i(\text{A})$
Device 1	15	-1
Device 2	5	1
Device 3	10	2
Device 4	-10	-1
Device 5	20	-3
Device 6	20	2

Find the power associated with each device and state whether the device is absorbing or delivering power. Use the power balance to check your work.

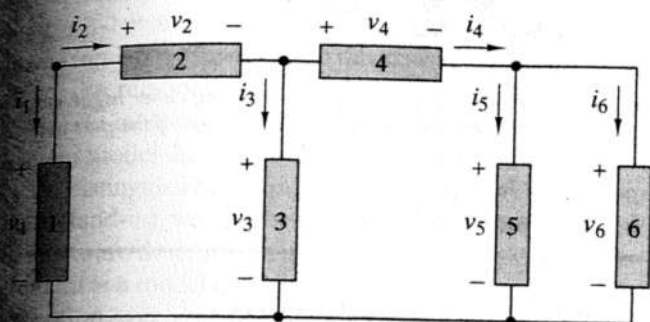


FIGURE P1-22

1-27 AC to DC Converter

A manufacturer's data sheet for the converter in Figure P1-27 states that the output voltage is $v_{dc} = 5 \text{ V}$ when the input voltage $v_{ac} = 120 \text{ V}$. When the load draws a current $i_{dc} = 40 \text{ A}$, the input power $p_{ac} = 300 \text{ W}$. Find the efficiency of the converter.

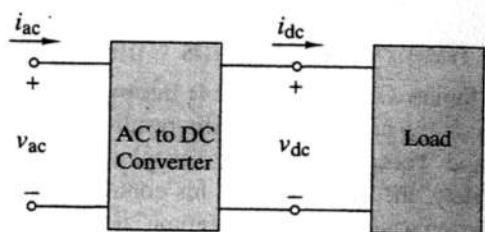


FIGURE P1-27

2-4 In Figure P2-4 the resistor dissipates 25 mW. Find R_x .

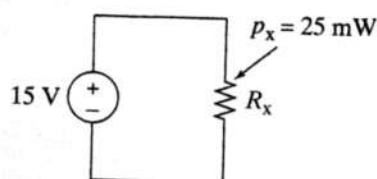


FIGURE P2-4

2-7 A $10\text{-k}\Omega$ resistor has a power rating of 0.25 W. Find the maximum voltage that can be applied to the resistor.

2-11 For the circuit in Figure P2-11:

- Identify the nodes and at least two loops.
- Identify any elements connected in series or in parallel.
- Write KCL and KVL connection equations for the circuit.

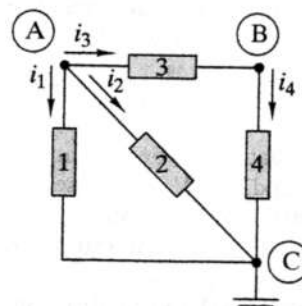


FIGURE P2-11

2-12 In Figure P2-11 $i_2 = 10 \text{ mA}$ and $i_4 = 20 \text{ mA}$. Find i_1 and i_3 .

2-21 Find v_x and i_x in Figure P2-21.

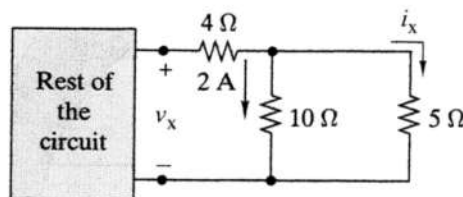


FIGURE P2-21

2-27 Find the equivalent resistance R_{EQ} in Figure P2-27.

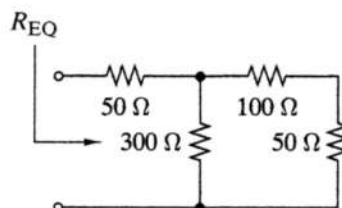


FIGURE P2-27

2-38 Select the value of R in Figure P2-38 so that $R_{AB} = R_L$.

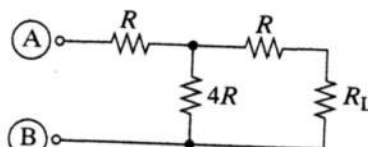


FIGURE P2-38